Karnaugh Map (K-Map) Simplification

Digital Logic Design

K-map

“When Boolean expressions get big, circuits need more gates $\longrightarrow$ more cost, more delay. We need a **shortcut method to simplify logic**.”

That shortcut = **Karnaugh Map (K-Map)**

It is just a **smart table** to reduce Boolean expressions visually instead of using long algebra.

K- Map

- **“Truth table tells WHAT the output is.
  K-Map tells HOW to make the circuit simple.”**

What is a Karnaugh Map?

“Truth table gives output… K-Map gives simplified circuit.”

- A graphical method to simplify Boolean expressions
- Reduces logic circuits
- Minimizes number of gates required
- Also called K-Map

Why Do We Use K-Maps?

- Simplifies complex Boolean expressions
- Helps in minimizing SOP and POS forms
- Reduces hardware cost
- Improves circuit reliability

K-Map for 2 Variables

- Contains 4 cells
- Variables: A and B
- Each cell represents a minterm
- Adjacent cells differ by only one variable

“If a variable changes inside a group → it disappears.”

For 2 variables A and B

AB	**0**	**1**
0	m0	m1
1	m2	m3

Each box = **one minterm**

Example:
F(A,B) = Σ(1,3)

Put 1 in cells **m1 and m3**

Now group them 👇

- m1 (01)
- m3 (11)

They differ only in **A**
B is always **1**

✅ Simplified Answer: **F = B**

K-Map for 3 Variables

- Contains 8 cells
- Variables: A, B, C
- Arranged in Gray code order
- Used for simplifying 3-variable expressions

Now variables: **A, B, C**

We arrange in **Gray Code order** (only one bit changes)

```makefile
          BC
         00   01   11   10
A=0      m0   m1   m3   m2
A=1      m4   m5   m7   m6
```

Example:

F = Σ(1,3,5,7)

All cells where **C = 1**

Group all four → result:

✅ **F = C**

K-Map for 4 Variables

- Contains 16 cells
- Variables: A, B, C, D
- Cells arranged using Gray code sequence
- Most commonly used K-Map in exams

Rules for Grouping

- **Step 5: Rules for Grouping (VERY IMPORTANT)**
- Write these rules clearly on board:
- 1 Group only **1s** (for SOP)
  2 Group size must be **power of 2** → 1, 2, 4, 8
  3 Make **largest possible group**
  4 Groups can **wrap around edges**
  5 A cell can be used in **multiple groups**

"Bigger group = simpler answer"

Example: SOP Simplification

- Given: $F(A,B,C) = \Sigma(1,3,5,7)$
- Plot 1s in corresponding cells
- Form groups of 2 or 4
- Write simplified expression

Example: POS Simplification

- Given: $F(A,B,C) = \Pi(0,2,4,6)$
- Plot 0s in corresponding cells
- Group zeros
- Write simplified POS expression

Advantages of K-Maps

- Easy visual method
- Reduces design complexity
- Time-saving for small variables
- Widely used in digital electronics

Limitations of K-Maps

- Not suitable for more than 4-5 variables
- Becomes complex for large expressions
- Computer algorithms preferred for many variables

Practice Problem

- Simplify $F(A,B,C,D) = \Sigma(0,2,5,7,8,10,13,15)$
- Step 1: Draw 4-variable K-Map
- Step 2: Plot 1s
- Step 3: Make largest groups
- Step 4: Write simplified expression